COMP 421 - Database Systems
Project 2: Database Creation

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1 Relational Model

The relational model has not been modified from Project 1 with the exception of replacing `date` and `time` columns in the `Note` table by `timestamp`.

Model

`HealthcareInstitution(hcId, phoneNum, name, email, address, website)`

`CommunityClinic(hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

`BirthingCenter(hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

`Midwife(pracId, name, email, phoneNum, hcId)`

`hcId` foreign key referencing `HealthcareInstitution`

note: `hcId` has to be `NOT NULL` to enforce Midwife's participation constraint in the `employs` relationship set.

`Father(fatherId, phoneNum, name, email, address, QChcardid, dateOfBirth, profession, bloodType)`

`Mother(QChcardid, phoneNum, name, email, adddress, dateOfBirth, profession, bloodType)`

`Couple(coupleId, invited, interested, selected, QChcardid, fatherId)`

`QChcardid` foreign key referencing `Mother`
`fatherId` foreign key referencing `Father`

note: `QChcardid` has to be `NOT NULL` to enforce Couple's participation constraint in the `constitutesF` relationship set.

`InformationSession(sessId, language, date, time, pracId)`

pracId foreign key referencing Midwife

note: pracId has to be NOT NULL to enforce InformationSession's participation constraint in the hosts relationship set.

Invitation(sessId, coupleId, attended, registered)

sessId foreign key referencing Information Session
coupleId foreign key referencing Couple

Pregnancy(coupleId, pregNum, estDueDate, expBirthTimeframe, lastMenstPeriod, isHomeBirth, primaryMidwife, backupMidwife)

coupleId foreign key referencing Couple
primaryMidwife foreign key referencing Midwife(pracId)
backupMidwife foreign key referencing Midwife(pracId)

note: coupleId is part of the primary key and thus has to be NOT NULL.

Appointment(apptId, date, time, pregNum, coupleId, pracId)

(pregNum, coupleId) foreign key referencing Pregnancy
pracId foreign key referencing Midwife

note: pregNum and coupleId has to be NOT NULL to enforce Appointment's participation constraint in the belongs relationship set.

note: pracId has to be NOT NULL to enforce Appointment's participation constraint in the sets up relationship set.

Note(noteId, timestamp, description, apptId)

apptId foreign key referencing Appointment

note: apptId has to be NOT NULL to enforce Note's participation constraint in the belongs relationship set.

Baby(babyId, name, gender, timeOfBirth, dateOfBirth, bloodType, pregNum, coupleId)

(pregNum, coupleId) foreign key referencing Pregnancy

note: pregNum and coupleId has to be NOT NULL to enforce Baby's participation constraint in the born from relationship set.

Technician(techId, name, phoneNum)

Test(testId, type, datePrescribed, dateSampleTaken, results, dateConducted, lab, techId, pracId, pregNum, coupleId, babyId, QChcardid)

techId foreign key referencing Technician
pracId foreign key referencing Midwife
(pregNum, coupleId) foreign key referencing Pregnancy
babyId foreign key referencing Baby

QChcardid foreign key referencing Mother

note: pregNum, coupleId has to be NOT NULL to enforce Test's participation constraint in the belongs relationship set.

note: techId has to be NOT NULL to enforce Test's participation constraint in the conducts relationship set.

note: pracId has to be NOT NULL to enforce Test's participation constraint in the prescribes relationship set.

BirthLocation(hcId, pregNum, coupleId) hcId foreign key referencing BirthingCenter

(pregNum, coupleId) foreign key referencing Pregnancy

Note: relation names have been pluralized in the db tables to follow convention.

Artificial keys

Entity Set	Artificial key
HealthcareInstitution	hcId
Father	fatherId
InformationSession	sessId
Test	testId
Baby	babyId
Appointment	apptId
Note	noteId
Couple	coupleId

Table 1: Artificial keys created in the relational model

2 Pending constraints

- The HealthcareInstitution, BirthingCenter, CommunityClinic ISA hierarchy is **disjoint** and **covering**. This cannot be reflected by the ER model or the relational translation and needs to be enforced by the application. Thus we will need to check that a record inserted into HealthcareInstitutions does not have foreign keys in both BirthingCenters and CommunityClinics and that every record in HealthcareInstitution is being referenced in child tables.
- The ER model and relational translation does not enforce the creation of exactly 1 couple per Father and Mother. In other words, the same set of Mother and Father can create multiple couples since a new artificial key is created for Couple. A check that a couple with the same parents don't exist should be done by the application before couple creation.
- Father is given an artificial key since unique attributes (i.e., email and Quebec Health Card Id) are optional for entity creation. The ER model cannot enforce the uniqueness of a father in the database if a father decides to register again since he will be given a new artificial key. Appropriate checks have to be done in the application. Furthermore, our database schemas do not verify the uniqueness of father's Quebec Health Card Id and email since these fields are optional and can be set to NULL.

- We keep track of a `Couple`'s pregnancy, not the mother. This means a new pregnancy with another man will be considered the couple's first pregnancy even though it is not the mother's first pregnancy.
- We cannot enforce the `Couple`'s participation constraint in the `belongs` relationship set with `Pregnancy` (i.e., the fact that a couple in the system always has a pregnancy) in the relational model. This should be enforced by the application.
- Appointments are set by the primary or backup midwife. To save which of the two midwives set up the `Appointment`, we created a relationship set (named `sets up`) between `Appointment` and `Midwife`. This cycle can cause inconsistencies in the database since the relational or ER models do not enforce the fact that an pregnancy's appointments can only be set up by the midwives responsible for that pregnancy. Appropriate checks need to be implemented by the application. The same applies to an appointment's `Notes` since we assume that the midwife setting up the appointment takes the note.
- Similar to the previous restrictions, due to cycles (relationship sets: `testsMother`, `testsBaby`, `prescribes`), the model does not enforce the test for a pregnancy be prescribed by the pregancy's primary or backup midwife. Moreover, the model does not enforce the test to be conducted on a baby or mother associated to the pregnancy. The `prescribes` relationship set was created to clarify which of the two midwives prescibed the tests, causing to this possible inconsistency. We could have instead added an attribute specifying which of the two midwives prescribed the test and let the application trace back to the midwife from the associated pregnancy. A cycle for `testsBaby` was however inevitable since a pregnancy can have multiple babies but evitable for mother since we can trace back to the mother from a pregnancy (a pregnancy only has one mother). Checks by the application have to be made to verify data consistency before inserting data.
- We cannot enforce in the relational model that a mother and father can only exist in the system if they are part of at least 1 couple. This participation constraint has to be enforced by our application.
- The relational model and db schemas fail to check all constraints across tables. For example, we do not check that the timestamp of a note belonging to an appointment has the same date as the appointment or if the birth date of a baby is later than the mother, etc.
- The model does not enforce that a Father minimally include their name, date of birth, phone number and profession. These attributes have to be `NOT NULL` during schemas definitions. This also applies to many other attributes in other entity sets.
- The model does not enforce that only parents that showed interest in the program and attended an information session and are selected will be assigned a midwife to their pregnancy. Furthermore, the model does not guarantee that a midwife will be associated to couples selected for the program.
- The model allows a midwife to be assigned to their own pregnancy if they are a mother or a father (since Midwife, Mother, and Father are 3 different entity sets).
- The system cannot enforce a female to register as a Father or a Male to register as mother. It is assumed that for example a user will register as Mother.
- Ideally, we would not want a Technician assigned to a pregnancy's test to be the father or mother of that pregnancy. We cannot enforce that since Technician is a different entity set than Father and Mother.
- It is mentioned in the project assumptions that emails can be considered to be unique to individuals. This is currently not enforced by the model since email is never a primary key (Mother: QChcardid, Father: fatherId, Midwife: pracId). Thus, a midiwfe can have the same email as another midwife or a mother/father can have the same email as another mother/father. We have to enforce this constraint in the application.
- Even though we have a basic email validation check constraint in the table creation for email, we could improve the email validation in the program.

3 SQL Queries

5 (a) - Write a SQL query that will list all the appointments for the midwife Marion Girard for the week March 21 - March 25 of this year (2022). The output must be the date and time of the appointment as well as the health care card, name and phone number of the mother. You can assume that there is only one midwife with this particular name.

```
SELECT date, time, Mothers.QChcardid, Mothers.name, Mothers.phoneNum
FROM Appointments
JOIN Midwives ON Midwives.pracId = Appointments.pracId
JOIN Couples ON Couples.coupleId = Appointments.coupleId
JOIN Mothers ON Mothers.QChcardid = Couples.QChcardid
WHERE Midwives.name = 'Marion Girard'
AND date BETWEEN '2022-03-21' AND '2022-03-25';
```

5 (b) - Write a SQL query that will list the results of all the blood iron tests that was performed on (Mother) Victoria Gutierrez for her second pregnancy. The output should have lab date and the result (the latter you can assume as a simple VARCHAR field with some text info - keep it a small field so your screen does not get cluttered).

```
WITH motherid (QChcardid) AS (
    SELECT QChcardid
    FROM Mothers
    WHERE name = 'Victoria Gutierrez'
)
SELECT dateConducted, results
FROM Tests
WHERE type = 'blood iron'
    AND (coupleId, pregNum) IN (
    SELECT coupleId, pregNum
    FROM Pregnancies
    WHERE coupleId = (
        SELECT coupleId
        FROM Couples
        WHERE QChcardid = (SELECT QChcardid FROM motherid)
    )
        AND pregNum = 2
)
    AND QChcardid = (SELECT QChcardid FROM motherid);
```

5 (c) - Write a SQL query that will list the names of each birthing center/community clinic as well as the number of pregnancies that are due for the month of July this year (2022). For this purpose you will count a pregnancy towards a facility if the primary midwife is from that facility (and not based on the final birthing location). You should also use the "final agreed up on" due date if that is present, in order to figure out if the pregnancy will fall into a specific month. Otherwise (if that information is not yet available), use the expected time frame for birth (year-month) that was provided initially.

```
SELECT HealthcareInstitutions.name,
    COALESCE(COUNT(primaryMidwife), 0)
        AS pregCount
FROM (
      SELECT *
      FROM Pregnancies
               JOIN Midwives ON Pregnancies.primaryMidwife = Midwives.
   pracId
      WHERE (finalEstDueDate BETWEEN '2022-07-01' AND '2022-07-31')
         OR (finalEstDueDate IS NULL
          AND YEAR(expBirthTimeframe) = 2022
          AND MONTH(expBirthTimeframe) = 7)
  ) pregnancies
      RIGHT OUTER JOIN HealthcareInstitutions
                       ON pregnancies.hcId = HealthcareInstitutions.hcId
GROUP BY HealthcareInstitutions.name;
```

5 (d) - Write a SQL query that will list the health care card, name and phone number of the mothers who is currently pregnant and have not yet given birth and is under the care of a midwife employed by Lac-Saint-Louis. You can assume that there is only one facility by that name.

```
WITH stLouisMidwives (pracId) AS (
    SELECT pracId
    FROM Midwives
    WHERE hcId = (
        SELECT hcid
        FROM HealthcareInstitutions
        WHERE name = 'Lac-Saint-Louis'
    )
)
SELECT DISTINCT Mothers.QChcardid, name, phoneNum
FROM Pregnancies
            JOIN Couples ON Pregnancies.coupleId = Couples.coupleId
            JOIN Mothers on Couples.QChcardid = Mothers.QChcardid
WHERE primaryMidwife IN (SELECT * FROM stLouisMidwives)
    OR backupMidwife IN (SELECT * FROM stLouisMidwives)
    AND (Pregnancies.coupleId, Pregnancies.pregNum) NOT IN (
        SELECT coupleId, pregNum
        FROM Babies
        WHERE dateOfBirth IS NOT NULL
    );
```

5 (e) -Write a SQL query that will list the health care card, name of mothers who have had more than one baby in a single pregnancy. List them only once even if they had multiple pregnancies where they had more than one baby. Count the babies whether they are born or not (yet).

```sql
SELECT QChcardid, name
FROM Mothers
WHERE QChcardid IN (
    SELECT QChcardid
    FROM Couples
    WHERE coupleId IN (
        SELECT coupleId
        FROM Babies
        GROUP BY (coupleId, pregNum)
        HAVING COUNT(*) > 1
    )
);
```

Screenshots

```
db2 => -- Write a SQL query that will list all the appointments for the midwife Marion Girard for the week March
-- 21 - March 25 of this year (2022). The output must be the date and time of the appointment as well
-- as the health care card, name and phone number of the mother. You can assume that there is only one
-- midwife with this particular name.
SELECT date, time, Mothers.QChcardid, Mothers.name, Mothers.phoneNum
FROM Appointments
        JOIN Midwives ON Midwives.pracId = Appointments.pracId
        JOIN Couples ON Couples.coupleId = Appointments.coupleId
        JOIN Mothers ON Mothers.QChcardid = Couples.QChcardid
WHERE Midwives.name = 'Marion Girard'
  AND date BETWEEN '2022-03-21' AND '2022-03-25';db2 => db2 => db2 => db2 => db2 (cont.) => db2 (cont.) => db2 (cont.)
=> db2 (cont.) => db2 (cont.) => db2 (cont.) =>

DATE       TIME     QCHCARDID    NAME                           PHONENUM
---------- -------- ------------ ------------------------------ ----------------------
03/21/2022 10:35:00 CASL23451234 Corina Solis                   613-555-0108
03/23/2022 11:35:00 BOCH07562649 Bobbi Cherry                   (403)557-2590
03/24/2022 15:30:00 KASA12232567 Katie Salinas                  (418)561-2943
03/25/2022 17:00:00 KASA12232567 Katie Salinas                  (418)561-2943
03/21/2022 18:10:00 JULE09156234 Judith Leon                    (514)723-9434

  5 record(s) selected.
```

Figure 1: Q5(a)

```
db2 => -- Write a SQL query that will list the results of all the blood iron tests that was performed on (Mother)
-- Victoria Gutierrez for her second pregnancy. The output should have lab date and the result (the latter
-- you can assume as a simple VARCHAR field with some text info - keep it a small field so your screen does
-- not get cluttered).
WITH motherid (QChcardid) AS (
    SELECT QChcardid
    FROM Mothers
    WHERE name = 'Victoria Gutierrez'
)
SELECT dateConducted, results
FROM Tests
WHERE type = 'blood iron'
  AND (coupleId, pregNum) IN (
    SELECT coupleId, pregNum
    FROM Pregnancies
    WHERE coupleId = (
        SELECT coupleId
        FROM Couples
        WHERE QChcardid = (SELECT QChcardid FROM motherid)
    )
      AND pregNum = 2
)
  AND QChcardid = (SELECT QChcardid FROM motherid);db2 => db2 => db2 => db2 => db2 (cont.) => db2 (cont.) => db2 (cont.
) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont
.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) =>

DATECONDUCTED RESULTS
------------ ----------------------------------------------------------------------------------------------------
01/04/2020    112 mcg/dL
02/25/2020    97 mcg/dL
02/28/2020    134 mcg/dL
01/25/2020    75 mcg/dL
03/12/2020    86 mcg/dL

  5 record(s) selected.
```

Figure 2: Q5(b)

```
db2 => -- Write a SQL query that will list the names of each birthing center/community clinic as well as the number
-- of pregnancies that are due for the month of July this year (2022). For this purpose you will count a
-- pregnancy towards a facility if the primary midwife is from that facility (and not based on the final birthing
-- location). You should also use the “final agreed up on” due date if that is present, in order to figure out
-- if the pregnancy will fall into a specific month. Otherwise (if that information is not yet available), use
-- the expected time frame for birth (year-month) that was provided initially.
SELECT HealthcareInstitutions.name, COALESCE(COUNT(primaryMidwife), 0) AS pregCount
FROM (
        SELECT *
        FROM Pregnancies
                JOIN Midwives ON Pregnancies.primaryMidwife = Midwives.pracId
        WHERE (finalEstDueDate BETWEEN '2022-07-01' AND '2022-07-31')
           OR (finalEstDueDate IS NULL AND YEAR(expBirthTimeframe) = 2022 AND MONTH(expBirthTimefdb2 => rame) = 7)
     ) pregnancies
        RIGHT OUTER JOIN HealthcareInstitutions
                        ON pregnancies.hcId = HealthcareInstitutions.hcId
GROUP BY HealthcareInstitutions.name;db2 => db2 => db2 => db2 => db2 => db2 (cont.) => db2 (cont.) => db2 (cont.) => db
2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) =>

NAME                                               PREGCOUNT
-------------------------------------------------- -----------
CLSC de Côte-des-Neiges                                      1
Centre hospitalier de l'Université de Montréal               0
Hôpital Notre-Dame                                           0
Jewish General Hospital                                      0
Lac-Saint-Louis                                              0
McGill University Health Center                              2
Michael Garron Hospital                                      2
Mount Sinai Hospital                                         0
St. Mary's Hospital Center                                   0
Toronto General Hospital                                     0

  10 record(s) selected.
```

Figure 3: Q5(c)

```
db2 => -- Write a SQL query that will list the health care card, name and phone number of the mothers who is
-- currently pregnant and have not yet given birth and is under the care of a midwife employed by Lac-
-- Saint-Louis. You can assume that there is only one facility by that name.
WITH stLouisMidwives (pracId) AS (
    SELECT pracId
    FROM Midwives
    WHERE hcId = (
        SELECT hcid
        FROM HealthcareInstitutions
        WHERE name = 'Lac-Saint-Louis'
    )
)
SELECT DISTINCT Mothers.QChcardid, name, phonedb2 => Num
FROM Pregnancies
         JOIN Couples ON Pregnancies.coupleId = Couples.coupleId
         JOIN Mothers on Couples.QChcardid = Mothers.QChcardid
WHERE primaryMidwife IN (SELECT * FROM stLouisMidwives)
   OR backupMidwife IN (SELECT * FROM stLdb2 => ouisMidwives)
    AND (Pregnancies.coupleId, Pregnancies.pregNum) NOT IN (
        SELECT coupleId, pregNum
        FROM Babies
        WHERE dateOfBirth IS NOT NULL
    );
db2 => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (co
nt.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (c
ont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) =>
QCHCARDID    NAME                           PHONENUM
------------ ------------------------------ ----------------------
JAGA23412231 Janette Gamble                 613-555-0184
CASL23451234 Corina Solis                   613-555-0108
JULE09156234 Judith Leon                    (514)723-9434
KASA12232567 Katie Salinas                  (418)561-2943
TIBR08957842 Tiffany Brennan                613-555-0114
VIGU63128923 Victoria Gutierrez             613-162-0131

  6 record(s) selected.
```

Figure 4: Q5(d)

```
db2 => -- Write a SQL query that will list the health care card, name of mothers who have had more than one baby
-- in a single pregnancy. List them only once even if they had multiple pregnancies where they had more
-- than one baby. Count the babies whether they are born or not (yet).
SELECT QChcardid, name
FROM Mothers
WHERE QChcardid IN (
    SELECT QChcardid
    FROM Couples
    WHERE coupleId IN (
        SELECT coupleId
        FROM Babies
        GROUP BY (coupleId, pregNum)
        HAVING COUNT(*) > 1
    )
);db2 => db2 => db2 => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (c
ont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) =>

QCHCARDID    NAME
------------ ------------------------------
JAGA23412231 Janette Gamble
CASL23451234 Corina Solis
JULE09156234 Judith Leon
TIBR08957842 Tiffany Brennan
VIGU63128923 Victoria Gutierrez

  5 record(s) selected.
```

Figure 5: Q5(e)

4 Midwife Information

6 (a)

```
1     CREATE VIEW midwifeinfo
```

```
    (pracId,
     name,
     phoneNum,
     email,
     InstituteName,
     InstituteAddress)
AS
SELECT pracId,
m.name,
m.phoneNum,
m.email,
hInst.name,
hInst.address
FROM Midwives m
 JOIN HealthcareInstitutions hInst on m.hcId = hInst.hcId;
```

6 (b)

```
db2 =>     CREATE VIEW midwifeinfo
    (pracId,
     name,
     phoneNum,
     email,
     InstituteName,
     InstituteAddress)
AS
SELECT pracId,
m.name,
m.phoneNum,
m.email,
hInst.name,
hInst.address
FROM Midwives m
 JOIN HealthcareInstitutions hInst on m.hcId = hInst.hcId;
db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (
cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.) => db2 (cont.
) => db2 (cont.) => db2 (cont.) => DB20000I  The SQL command completed successfully.
```

Figure 6: midwifeinfo view creation

6 (c)

```
db2 => SELECT *
FROM midwifeinfo
LIMIT 5;db2 (cont.) => db2 (cont.) =>

PRACID      NAME                      PHONENUM          EMAIL                                    INSTITUTENAME                                 INSTITUTEADDRESS
----------- ------------------------- ----------------- ---------------------------------------- --------------------------------------------- ----------------------------------------------------------
----------
      1001 Marion Girard             514 556-6666      mgirard@muhc.mcgill.ca                   McGill University Health Center               1001 Decarie Blvd, Montreal, Quebec H4A 3J1
      1005 Rauf Pasternack           997 764-1123      raufpasternack@smhc.qc.ca                St. Mary's Hospital Center                    3830 Lacombe Avenue, Montreal, Quebec H3T 1M5
      1002 Baal Hammon Maly          675 662-9861      hammonm@stlouishospital.qc.ca            Lac-Saint-Louis                               180, avenue Cartier Pointe-Claire, Québec H9S 4S1
      1003 Najla Vazquez             221 121-7812      nvazquez@tehn.ca                         Michael Garron Hospital                       825 Coxwell Avenue Toronto ON M4C 3E7
      1004 Demeter Kralj             234 233-2234      demeterk@tehn.ca                         Michael Garron Hospital                       825 Coxwell Avenue Toronto ON M4C 3E7

  5 record(s) selected.
```

Figure 7: midwifeinfo view query

6 (d)

```
db2 => SELECT * FROM midwifeinfo WHERE INSTITUTENAME = 'Lac-Saint-Louis';

PRACID      NAME                          PHONENUM            EMAIL                                              INSTITUTENAME                                      INSTITUTEADDRESS
----------- ----------------------------- ------------------- -------------------------------------------------- -------------------------------------------------- --------------------------------------------------
----------
       1002 Baal Hammon Maly              675 662-9861        hammonm@stlouishospital.qc.ca                      Lac-Saint-Louis                                    180, avenue Cartier Pointe-Claire, Québec H9S 4S1

       1007 Nisha Harper                  514 898-6129        nisha.harper@muhc.mcgill.ca                        Lac-Saint-Louis                                    180, avenue Cartier Pointe-Claire, Québec H9S 4S1

  2 record(s) selected.
```

Figure 8: midwifeinfo view query where on midwives employed at Lac-Saint-Louis

6 (e) - Insering valid data into the view fails (as show in Figure 9). In theory, the reason behing this is that views are unmaterialized relations, storing a query definition rather than a set of tuples and thus cannot be treated as tables and tuples cannot be inserted into them.

Insering into views is allowed in IBM DB2 (it is not only a unmaterialized relation in the DB implementation). However, the `midwifeinfo` view is created of two base tables `Midwives` and `HealthcareInstitutions`, that have attributes not presented in the view with `NOT-NULL` constraints and no default values. Therefore, inserting in the view fails. Furthermore, DB2 documentation states that `JOIN`s in the view definition prohibit further updates to the view, which can be yet another reason this operation fails.

```
db2 => INSERT INTO midwifeinfo VALUES(100000, 'Ashley Wayne', '514 565-2232', 'ashley@gmail.com', 'Lac-Saint-Louis','180, avenue Cartier Pointe-Claire, Québec H9S 4S1');
DB21034E  The command was processed as an SQL statement because it was not a
valid Command Line Processor command.  During SQL processing it returned:
SQL0150N  The target fullselect, view, typed table, materialized query table,
range-clustered table, or staging table in the INSERT, DELETE, UPDATE, MERGE,
or TRUNCATE statement is a target for which the requested operation is not
permitted.  SQLSTATE=42807
db2 =>
```

Figure 9: view insertion

5 Check Constraints

```
db2 => ALTER TABLE Tests
    ADD CONSTRAINT dateconstraint CHECK (dateConducted >= datePrescribed);
db2 (cont.) => DB20000I  The SQL command completed successfully.
```

Figure 10: `CHECK` constraint creation

```
db2 => INSERT INTO Tests
VALUES (101021, 'blood', '2022-03-23', '2022-03-24', 'All good', '2022-02-24', 'McGill labs', 8500, 1004, 1, 4004,
        6905, NULL);db2 (cont.) => db2 (cont.) =>
DB21034E  The command was processed as an SQL statement because it was not a
valid Command Line Processor command.  During SQL processing it returned:
SQL0545N  The requested operation is not allowed because a row does not
satisfy the check constraint "SZINAT.TESTS.DATECONSTRAINT".  SQLSTATE=23513
```

Figure 11: Inserting data with `dateConducted` before `datePrescribed` fails due to the check constraint. Note: 2022-02-24 is the `dateConducted` and 2022-03-23 is the `datePrescribed`